

NICHOLAS YAP CHEE ANG

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<https://github.com/cyrolite>

EDUCATION

National University of Singapore

Aug 2023 - May 2027

Bachelor of Computing in Information Security

- Second Major in Mathematics
- Current Year 4 Undergraduate
- GPA: 4.54 / 5

SKILLS

- Cybersecurity: Network Security, Cryptography, Digital Forensics
- Cybersecurity Tools: BurpSuite, Wireshark, GDB
- Programming: Java, Python, C++, C, Rust
- Web Development: HTML, CSS
- Database Management: MySQL, PostgreSQL, SQLite
- Soft Skills: Leadership, Public Speaking, Problem Solving
- Statistical Analysis Tools: Radiant, Microsoft Excel, Tableau

PROJECTS

AI-Powered Local Penetration Testing Assistant

Feb 2026 - Present

- Built a local AI security assistant powered by a fine-tuned LLM, RAG over 250,000 CVEs, and security tool integration.
- Built a RAG pipeline over 250,000 real CVE records using ChromaDB and sentence transformers for accurate vulnerability lookups.
- Implemented voice interface using OpenAI Whisper (STT) and pyttsx3 (TTS) for hands-free security analysis.
- Deployed offline on local hardware with no cloud dependencies using Mistral.

Packet Sniffer

Dec 2024 - Present

- Created an application to sniff over 10,000 of packets, capturing Ethernet, IP, TCP, and UDP packet details in real-time.

CollabSync

Feb 2025 - May 2025

- Led a team of 4 to engage in a brownfield project of creating a functional address book for NUS students.
- Coordinated weekly meetings with team to ensure tasks are completed on time.

Scooby Chatbot

Jan 2025 - Feb 2025

- Created an off-line local task chatbot to assist user's with keeping track of over 100 of tasks.
- Wrote over 1000 lines of code, including test code as well as functional code.

PassManager

May 2024 - Aug 2024

- Developed an online web application (Website: <https://cyrolite.pythonanywhere.com/>) with a partner to store over 10,000 passwords with AES encryption, along with login / logout capabilities.

CO-CURRICULAR ACTIVITIES

STANDCON Capture the Flag (CTF), Member

Dec 2023 - Jan 2024

- Competed in a competitive CTF event, solving over 10 challenges in digital forensics, binary exploitation, and web vulnerabilities.

Acronis Capture the Flag (CTF) Competition, Team Leader

Dec 2023 - Dec 2023

- Led a team in tackling over 10 advanced cybersecurity challenges, including reverse engineering, cryptography, and network analysis.
- Divided tasks among 5 team members, resulting in efficient problem-solving and high team performance.

ADDITIONAL COURSES

- Computer Forensics Fundamentals
- The Complete SQL Bootcamp: Go from Zero To Hero
- Cryptography and Hashing Fundamentals in Python and Java

- Intro to Machine Learning
- Getting Started with AWS Services Fundamentals for Beginners

VOLUNTEER EXPERIENCE

Project Dawn, Leader

May 2024 - Present

- Starting up a volunteer organisation along with friends to raise awareness of cyber security.

Malay Activity Executive Committees Council, Mentor

Feb 2023

- Guided over 10 students every Sunday in Bukit Batok Community Centre.

EXPERIENCE

SmileTutor Pte. Ltd., Part-Time Tuition Teacher

Nov 2022 - Present

- Developed effective teaching methods for more than 5 students to better understand concepts.
- Acquired effective oratorical skills and effective note-taking skills for 2 students.
- Managed to help 10 students acquire a grade of B or higher for O Level Additional Mathematics, Mathematics and Physics.

Tiktok Private Limited, Cybersecurity Engineer Intern

Aug 2026 - Nov 2026

- Analysed HTTP request headers and behavioural features, including User-Agent, user identifiers, and device fingerprints, to characterise web-scraping patterns and attacker activity across TikTok Shop traffic at production scale (~1M requests/sec).
- Facilitated to development and validation of a multi-agent AI security system built with LangGraph, comprising detection and feature-analysis agents process security queries, identify relevant scraping patterns, and generate analytical responses.
- Collaborated with security and engineering teams to investigate emerging scraping behaviours and communicate findings informed defensive detection and mitigation strategies.

Infineon Technologies Asia Pacific Pte Ltd, Data Automation Engineer Intern

Dec 2025 - Jun 2026

- Architected and integrated a real-time alerting ecosystem featuring an automated email fault-detection system and custom Webex Bot notifications to instantly flag data run failures for over 20 scripts, flag pipeline anomalies and protect data availability, minimising operational data gaps.
- Deployed data logging telemetry directly into production data workflows, drastically reducing debugging times and enhancing long-term system auditability, support and data integrity.
- Leveraged Large Language Models (LLMs) to automate log analysis, engineering specialised prompt templates to ingest raw pipeline script errors and instantly generate actionable, human-readable troubleshooting summaries for over 20 different scripts.
- Engineered a predictive XGBoost machine learning model from scratch to automate root-cause exploration for electrical lot rejection rate (eLRR) spikes over 7 sites globally.
- Cooperated with global stakeholders to remodel and integrate new data to the SDT Data Quality Monitoring workbooks on Tableau dashboards.
- Implemented secure authentication protocols utilising REST API keys to facilitate authorised, programmatic communication between backend data sources and reporting infrastructure for over 10 dashboards.

Accenture Private Limited, Systems Deployment Facilitator

Jun 2025 - Jul 2025

- Expedited in rollout of National Billing System (NBS), a country-wide initiative to replace legacy hospital billing system in Singapore.
- Collaborated alongside developers and business analysts to triage and validate over 100 production bugs, ensuring quick turnaround for urgent fixes.
- Analysed discrepancies over 100 between expected vs. actual billing outcomes, contributing to early detection of major financial impact issues.
- Collaborated with stakeholders from over 20 healthcare clusters to consolidate bug reports, streamline resolution workflows, and improve system stability post-launch.

National University of Singapore, Undergraduate Teaching Assistant

Aug 2024 - Dec 2024

- Taught over 40 undergraduate students on basics of Computer Organisations.
- Marked over 40 undergraduate students' lab sheets for lab assignments.
- Demonstrated ability to guide 2 classes of undergraduates to solve problems in demonstrations or debug codes.
- Received an outstanding rating of 4.5 / 5 in terms of teaching.

ADDITIONAL INFORMATION

- English (Proficient)
- Chinese (Proficient)